

Solutions to Problem Set 3, Second Half

180.301 Microeconomic Theory - Spring 2026

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1. Using backward induction, we first solve the profit maximization problem of Firm B, treating y_A as given.

$p = 1000 - 0.5y = 1000 - 0.5y_A - 0.5y_B$, Total revenue $TR_B = (1000 - 0.5y_A - 0.5y_B)y_B$.
Marginal revenue $MR_B = 1000 - 0.5y_A - y_B$.

Profit maximization means $MR_B = MC_B$: $1000 - 0.5y_A - y_B = 4y_B \Rightarrow y_B = 200 - 0.1y_A$.

Now, knowing that Firm B will produce $y_B = 200 - 0.1y_A$, we solve the profit maximization problem of y_A .

For Firm A, $p = 1000 - 0.5y_A - 0.5y_B = 1000 - 0.5y_A - 0.5(200 - 0.1y_A) = 900 - 0.45y_A$.
 $MR_A = 900 - 0.9y_A$, $MR_A = MC_A$: $900 - 0.9y_A = 2y_A \Rightarrow y_A^* = \frac{9000}{29}$, $y_B^* = 200 - 0.1y_A^* = \frac{4900}{29}$.

2. In a price-leadership game, Firm A first chooses the price, and then Firm B takes this price as given and chooses its quantity.

Using backward induction, for Firm B: $p = 1000 - 0.5y_A - 0.5y_B$.

$0.5y_A = 1000 - p - 0.5y_B \Rightarrow y_A = 2000 - 2p - y_B$. Since Firm B treats price p as given, its profit maximization problem is $p = MC_B$: $p = 4y_B \Rightarrow y_B = \frac{p}{4}$.

Now, knowing that Firm B will produce $y_B = \frac{p}{4}$, we solve the profit maximization problem of Firm A.

$p = 1000 - 0.5y_A - 0.5y_B = 1000 - 0.5y_A - \frac{p}{8} \Rightarrow y_A = 2000 - \frac{9p}{4}$. Its profit is $\pi_A = TR_A - TC_A = p(2000 - \frac{9p}{4}) - (2000 - \frac{9p}{4})^2$.

The derivative with regard to p is $\frac{d\pi_A}{dp} = 2000 - \frac{9p}{2} - 2 \cdot (2000 - \frac{9p}{4}) \cdot (-\frac{9}{4}) = 11000 - \frac{117p}{8} = 0$. Solving, we get $p^* = \frac{88000}{117}$, $y_A^* = 2000 - \frac{9p}{4} = \frac{4000}{13}$, $y_B^* = \frac{p}{4} = \frac{22000}{117}$.

3. Treating Firm A's production y_A as given, Firm B's $MR_B = MC_B$: $y_B = 200 - 0.1y_A$.

Treating Firm B's production y_B as given, Firm A's $MR_A = 1000 - y_A - 0.5y_B$, $MC_A = 2y_A$, so $MR_A = MC_A \Rightarrow y_A = \frac{1000 - 0.5y_B}{3}$.

Solving the two equations in the preceding two paragraphs, we get the equilibrium $y_A^* = \frac{18000}{59}$, $y_B^* = \frac{10000}{59}$.

4. The Bertrand equilibrium has both firms setting the same prices equal to their marginal costs: $p = MC_A = MC_B \Rightarrow p = 2y_A = 4y_B$. With this, we can express y_A and y_B in terms of p : $y_A = \frac{p}{2}$, $y_B = \frac{p}{4}$.

The demand curve is $p = 1000 - 0.5y_A - 0.5y_B = 1000 - \frac{3p}{8} \Rightarrow p_A^* = p_B^* = \frac{8000}{11}$, $y_A^* = \frac{p}{2} = \frac{4000}{11}$, $y_B^* = \frac{p}{4} = \frac{2000}{11}$.

Figure 1 illustrates this equilibrium. In the “Market” graph, the market supply of $p = \frac{4}{3}y$ is derived by horizontally summing the MC curves of the two firms: Firm A’s supply curve is $p = 2y_A \Rightarrow y_A = \frac{p}{2}$; Firm B’s supply curve is $p = 4y_B \Rightarrow y_B = \frac{p}{4}$, so market supply is $y = y_A + y_B = \frac{p}{2} + \frac{p}{4} = \frac{3}{4}p \Rightarrow p = \frac{4}{3}y$.

As we can see from the graph, no firm can increase its profit by deviating from this equilibrium.

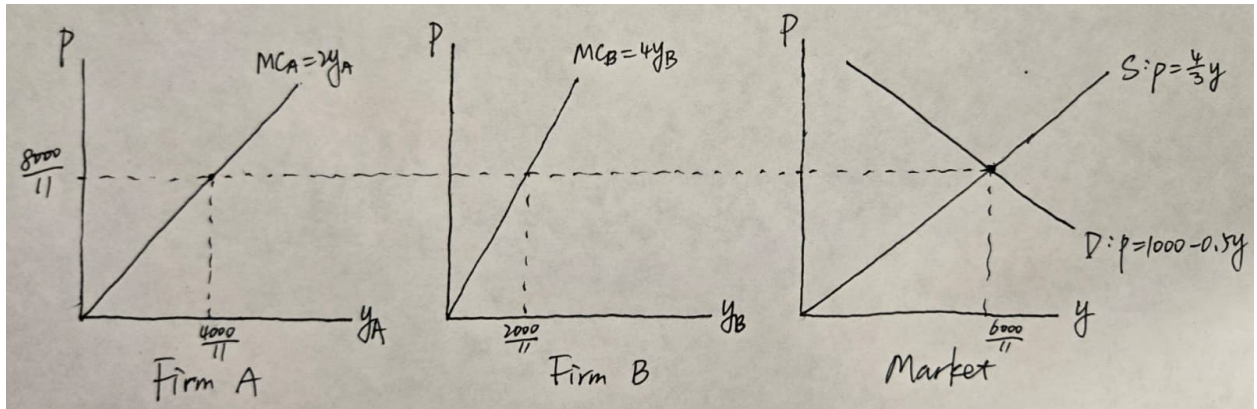


Figure 1: The Bertrand equilibrium

We now show that other prices cannot be an equilibrium. Suppose they set prices higher than marginal cost. Figure 2 illustrates an example where $p_A = p_B > MC$. At this price, market demand is smaller than market supply, so the firms can only jointly supply a quantity of y to the market (see the third graph of Figure 2).

Then, Firm A has an incentive to deviate from this price by setting a price of $p'_A = p - \epsilon$ (with ϵ small). Now, every consumer in the market will buy from Firm A. Firm A’s quantity increases from y_A to y'_A in the first graph of Figure 2, whereas Firm B’s quantity is 0. Firm A earns additional profit equal to the red shaded area from the new sales, and loses profit equal to the black shaded area from the lower price of existing sales. Overall, Firm A’s profit will increase from this deviation. So, a price that is higher than MC cannot be an equilibrium.

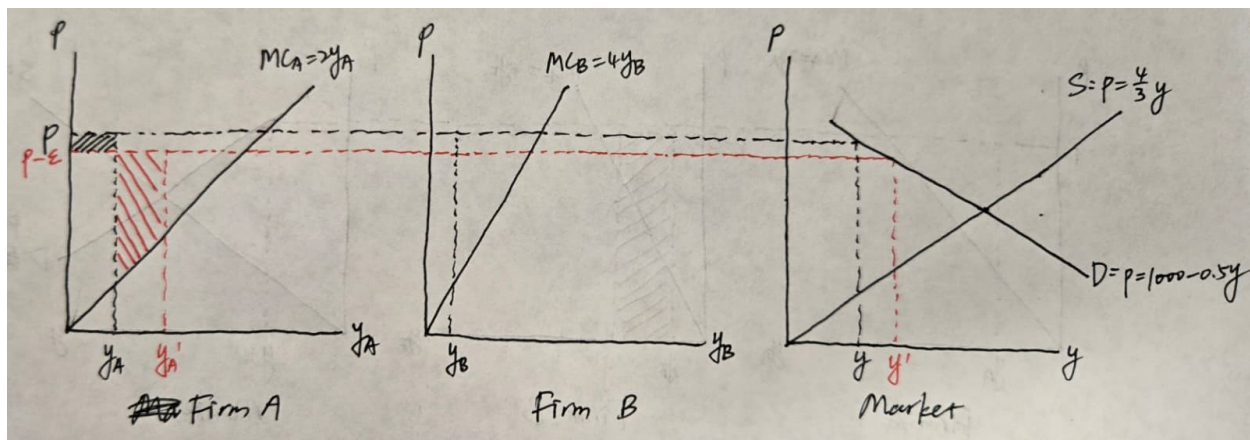


Figure 2: Prices higher than MC cannot be an equilibrium

Setting prices lower than marginal cost cannot be an equilibrium either. Figure 3 graphically illustrates this. At the price p , market demand exceeds supply, so the firms' quantities are constrained by supply. Now, Firm A can profitably deviate by increasing its price a bit to $p + \epsilon$ (with ϵ small). Because demand still exceeds supply, some consumers will buy from Firm A even if its price is now higher than Firm B. Firm A earns additional profit equal to the red shaded area. Therefore, a price that is lower than MC cannot be an equilibrium.

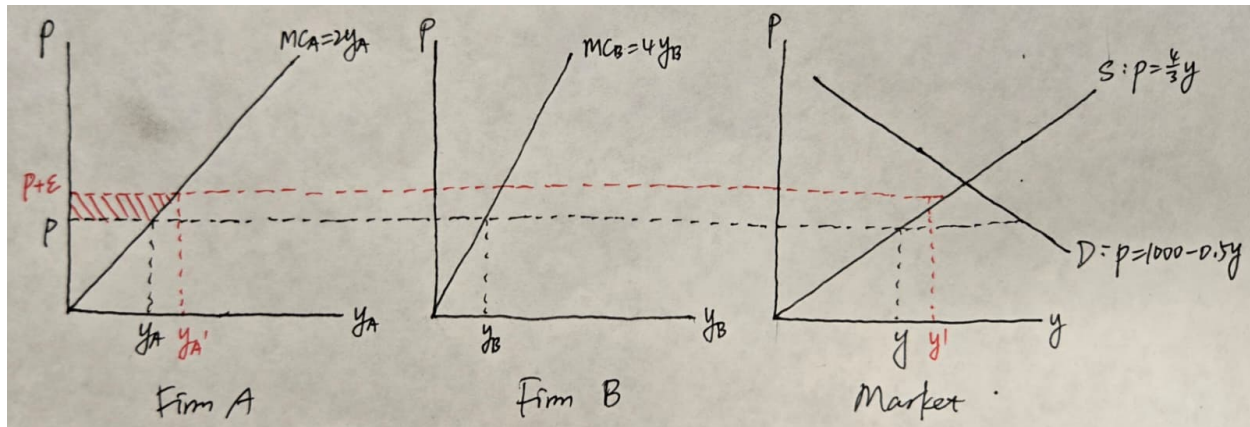


Figure 3: Prices lower than MC cannot be an equilibrium

Using similar arguments, we can also show that $p_A \neq p_B$ cannot be an equilibrium.

5. When the firms form a cartel, they can cooperate and maximize their joint profit.

First, notice that in a cartel, we must have $MC_A = MC_B$, otherwise production can be shifted from the firm with higher MC to the firm with lower MC to increase profit.

$$\text{So, } MC_A = MC_B \Rightarrow 2y_A = 4y_B \Rightarrow y_A = 2y_B.$$

$$\text{Demand curve: } p = 1000 - 0.5y_A - 0.5y_B = 1000 - 1.5y_B.$$

Total profit $\pi = p(y_A + y_B) - y_A^2 - 2y_B^2 = (1000 - 1.5y_B)(2y_B + y_B) - (2y_B)^2 - 2y_B^2 = 3000y_B - 10.5y_B^2$.

Taking derivative with regard to y_B : $\frac{d\pi}{dy_B} = 3000 - 21y_B = 0 \Rightarrow y_B^* = \frac{1000}{7}$, $y_A^* = \frac{2000}{7}$, $p^* = \frac{5500}{7}$.